INSTITUTE OF SPACE TECHNOLOGY
KICSIT, Kahuta Campus

Project Report

Advanced Visual Programming Lab

ONLINE SHOPPING MANAGEMENT SYSTEM

Console-Based Application using C# and SQLite

Submitted By

Laiba Naeem (232201058)
BSCS-6-B

Submitted To

Sir Uzair

Submission Date

April 15, 2026

Abstract

This project is a console-based Online Shopping Management System developed using C# and SQLite. It simulates a real-world e-commerce system with Admin and Customer roles. The system includes product management, cart handling, order processing, payment simulation, and review system.

1 Introduction

Online shopping systems are widely used in modern applications. This project demonstrates how an e-commerce system works using a console interface and SQLite database integration. It focuses on modular programming and backend logic.

2 Objectives

- Develop a console-based e-commerce system
- Implement Admin and Customer modules
- Manage products, cart, and orders
- Integrate SQLite database
- Simulate payment system

3 System Features

3.1 Admin Features

- Add, Update, Delete Products
- View All Products
- Manage Stock
- View Orders
- View Reviews

3.2 Customer Features

- Browse Products
- Search Products
- Add to Cart
- Checkout System
- Payment Options

- Order History
- Write Reviews

4 System Architecture

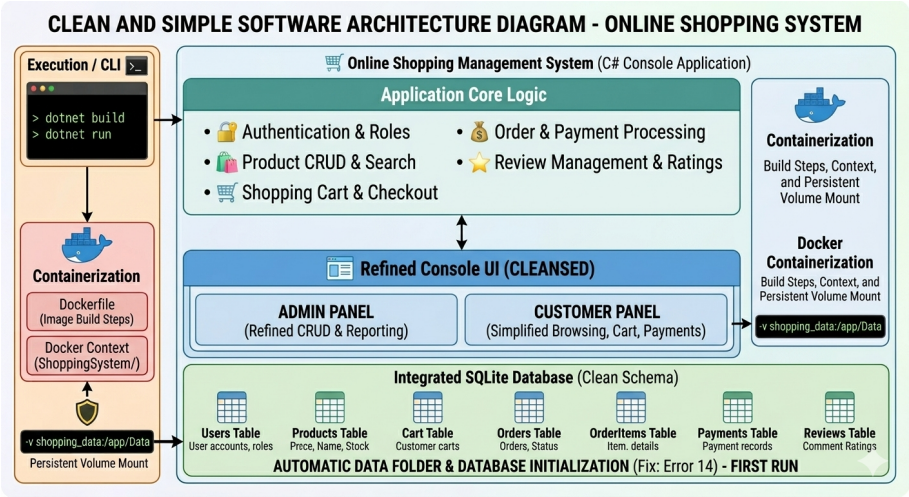

Figure 1: System Architecture Diagram

5 Entity Relationship Diagram (ERD)

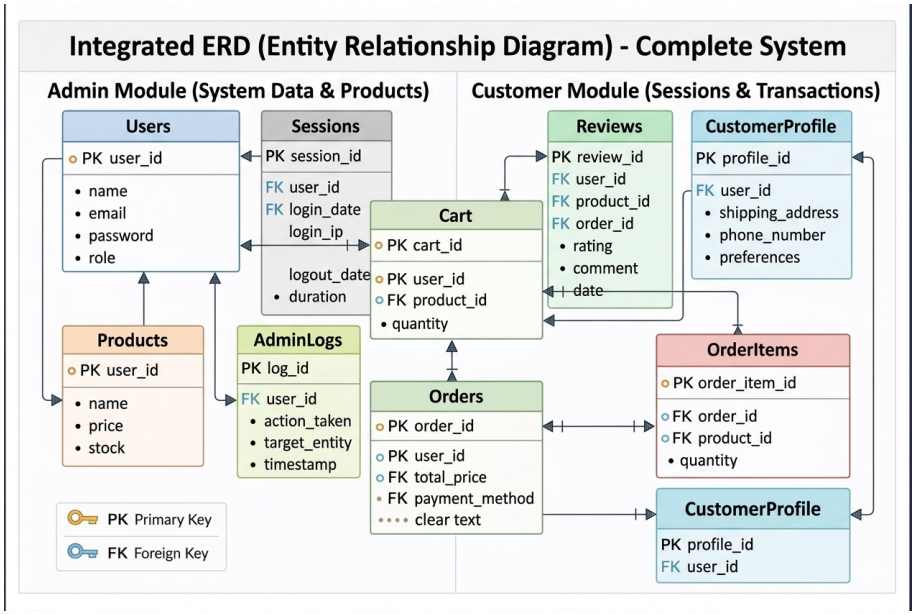

Figure 2: Database ERD

6 System Flow

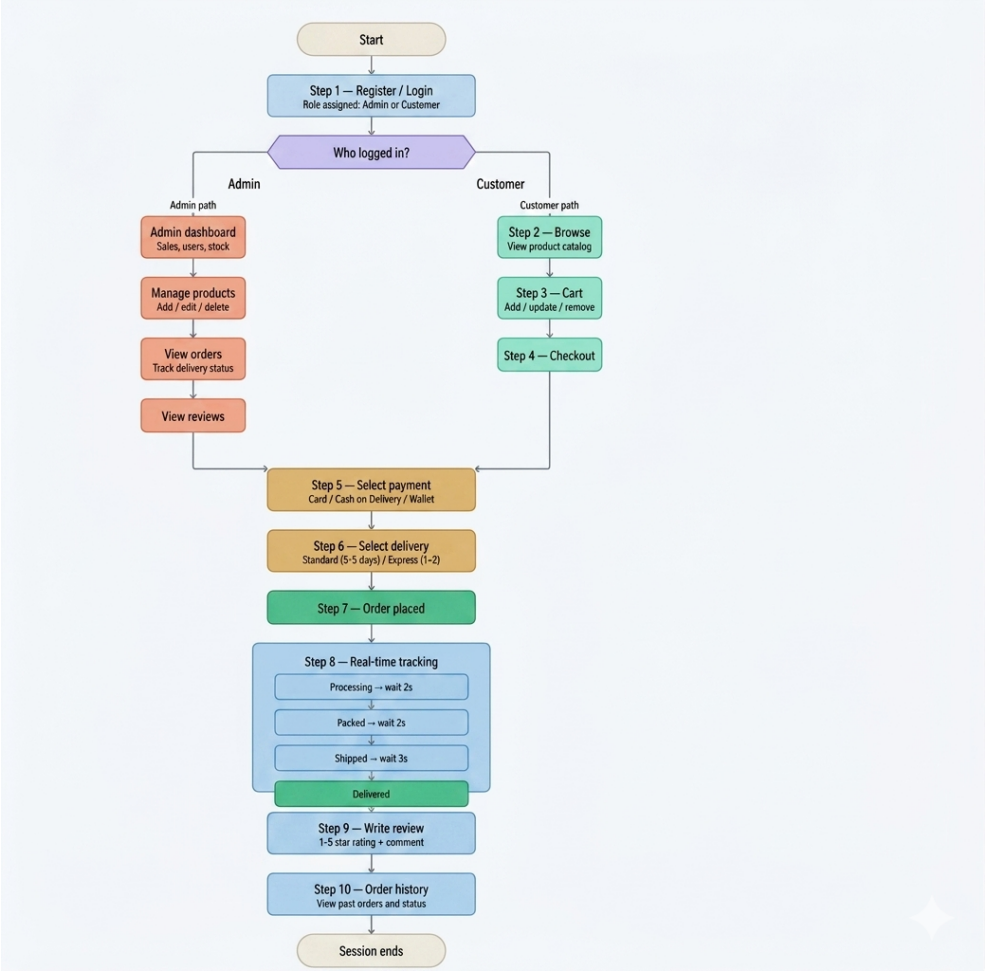

Figure 3: System Flow Diagram

7 Database Design

- Users Table (Admin & Customers)
- Products Table
- Cart Table
- Orders Table
- OrderItems Table
- Payments Table
- Reviews Table

8 Sample Code

8.1 SQLite Connection

```
using Microsoft.Data.Sqlite;

string connectionString = "Data Source=shopping.db";

using var connection = new SqliteConnection(connectionString);
connection.Open();
```

9 Main Interface

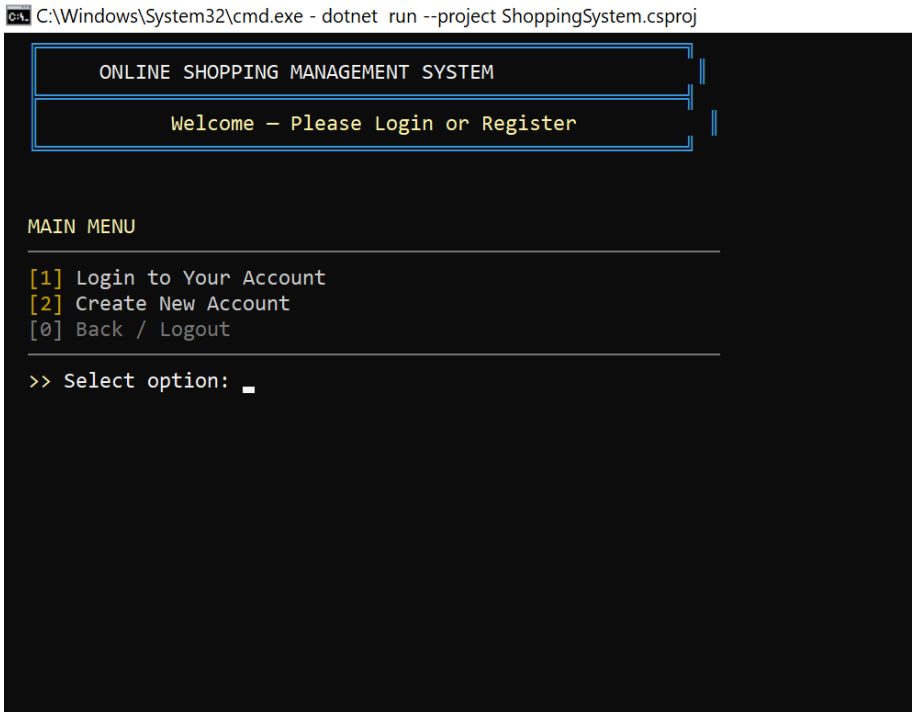

Figure 4: Main Menu Interface

10 Admin u Interface

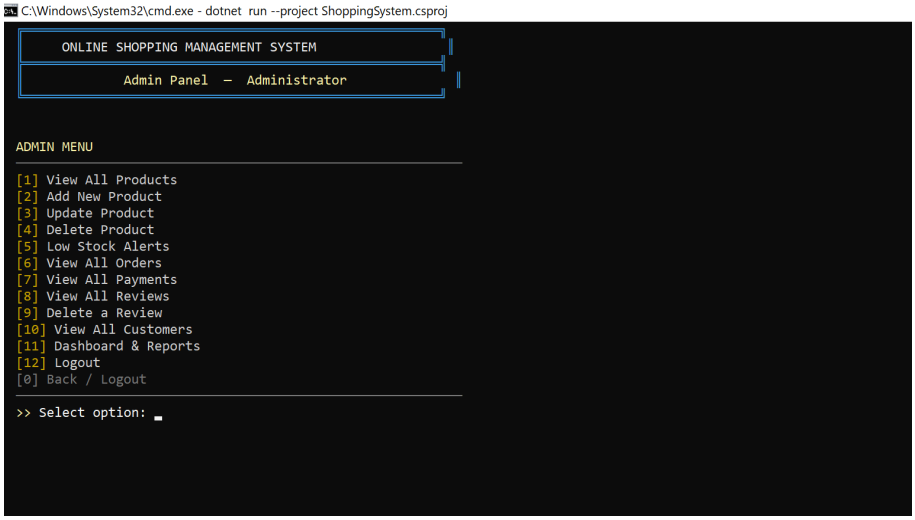

Figure 5: Admin u Interface

11 Customer Interface

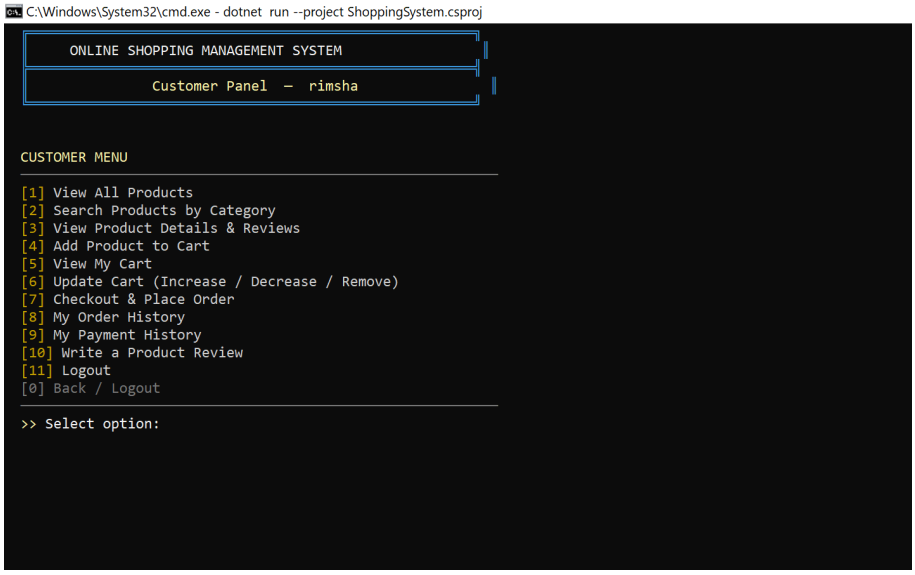

Figure 6: Customer Interface

12 Conclusion

The project successfully demonstrates a console-based e-commerce system using C# and SQLite. It includes real-world features like product management, cart system, and order processing. This project can be extended into a web or desktop application in the future.